MIT SCHOOL OF COMPUTING
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MIT-ADT UNIVERSITY
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A Leap Towards World Class Education

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Subject: PL Roll No.: 01 Exp. No.: 01
Name of the Experiment: Assignment-1

Marks	Teacher's Signature with date
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Performed on:
Submitted on:

1. List the Key features and advantages of Python. Also mention any two Python IDEs or editors used for program development.

→ Key features of Python :-

1) Simple and Easy to learn - Uses simple English-like syntax.
2) Interpreted language - Code is executed line by line.
3) High-level language - No need to manage memory manually.
4) Object-oriented - Supports classes and objects.
5) Portable - Same program run on windows, Linux, Mac.
6) Large library support - Has buit-in modules for many task.

Advantages of Python :-

1) Faster program development
2) Fewer lines in code
3) Used in web development, data science, AI, ML, automation

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

IV) Free and open-source

Python IDEs and editors
I) ~~I~~DLE - Default python IDE, simple and beginner friendly.
II) VS code - Popular editor with extension for python.

2. Explain the history and evolution of python? Why is python popular?
→ History and Evolution of Python :-
I) Python was developed by "Guido Van Rossum" in 1991.
II) Named after the TV show "Monty Python's Flying Circus."
III) It was designed to be simple, readable & powerful.
IV) Major version : Python 2.x (older), Python 3.x (current & improved).

Python is popular because :-
I) It is easy for beginners
II) It is used in industries (Google, Netflix, Nasa, etc).
III) It supports many fields : AI, Data science, Web development.
IV) It has large community support.

3. Write a Python program to accept two numeric values from the user and perform the following operation : Addition, Substraction, Multiplication, Division
Display the result using proper input & output programing.

→
```
a = float (input ("Enter first number : "))
b = float (input ("Enter ascend number : "))
print ("Addition : ", a+b)
print ("Substraction : ", a-b)
print ("Multiplication : ", a*b)
print ("Division : ", a/b)
```

4. Analyze the following Python code and identify the data types of such variable. Also explain the output.

1) a = 10
2) b = 5.5
3) c = 3 + 4j
4) d = True
5) print (a+b, type (c), d)

→ Data types

Variable	Values	Data type
a	10	Integer
b	5.5	Float
c	3+4j	Complex
d	True	Boolean

Output :- 15.5, < class 'complex' > True

Explation :-

a+b = 10 + 5.5 = 15.5

type (c) show complex data types

d print True

5. Evaluate the use of relational, logical and bitwise operators in Python. Justify with suitable example where each type of operator is most appropriate?

→ ~~Refa~~

1) Relational operator :- Used to compared value

```
a = 10          a = 10
b = 5           b = 5
print (a>b)     print (a<b)
Output : True   Output : False
```

11) Logical Operator :- Used to combine conditions

```
x = True            x = True
y = False           y = True
print (x and y)     print (x and y)
output : False      Output : True
```

111) Bitwise operator :- Used at bit level

```
a = 5      # 0101
b = 9      # 0011
print (a&b)
output : 1
```

6. Design and implement a menu-driven Python program that : Accept user input, Perform arithmetic, relational, logical and assignment ~~to~~ operation, Display result clearly.

The program should handle int, float and boolean data type.

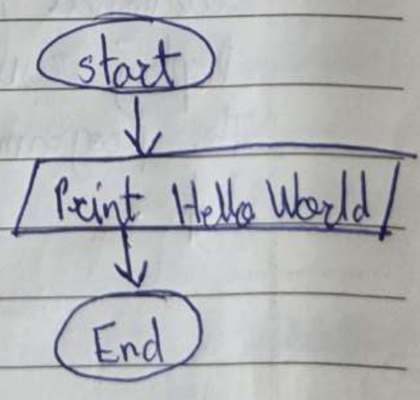
```
→ a = int (input ("Enter First number : "))
  b= int (input (" Enter second number : "))
  print ("1. Addition ")
  print (" 2. Greater than check ")
  print (" 3. logical AND ")
  print ("4. Assignment operation ")
  choice = int (input ("Enter your choic "))
  if choice ==1 :
          print (" Addition : " , a+b)
  else if choice == 2 :
          print (" a > b : ", a > b)
  else if choice == 3 :
          print (" Logical AND : " , a >0 and b ≥ 0).
  else if choice == 4 :
          a+ = b
          print (" After assignment , a= " , a)
  else
          print (" Invalid choice ").
```

7\. Write a program in python to display " Hello World"

```
→ print (" Hello World ! ")
```

output :- Hello World !

Flowchart :-



Algorithm :-

1. Start
2. Print the message " Hello World "
3. Stop

8. Write a program code to perform average of 5 number using () function and type conversion concept.

→ Code :-

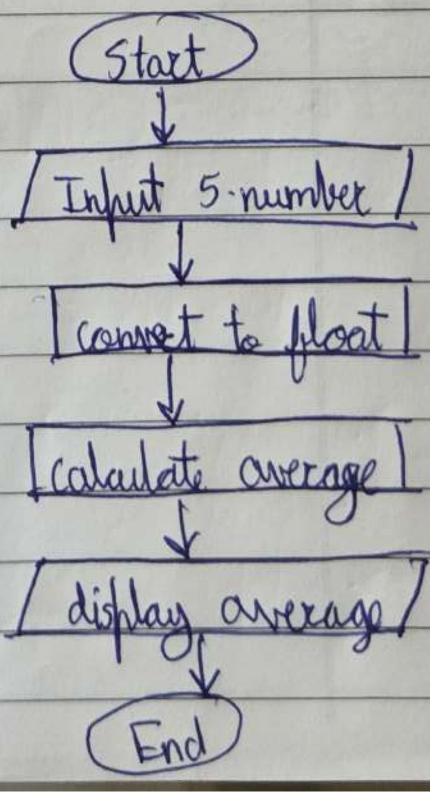
```
a = float (input ("Enter number 1 : "))
b = float (input ("Enter number 2 : "))
c = float (input ("Enter number 3 : "))
d = float (input ("Enter number 4 : "))
e = float (input ("Enter number 5 : "))

avg = (a+b+c+d+e)
          5

print ("Avg : " , avg )
```

Output :-

Enter number 1 = 1
Enter number 2 = 5
Enter number 3 = 3
Enter number 4 = 1
Enter number 5 = 5
Avg = 3

* Algorithm :
1. start
2. Read five numbers using input ()
3. Convert inputs to float
4. Calculate average
5. Display average
6. End

* Flowchart :-

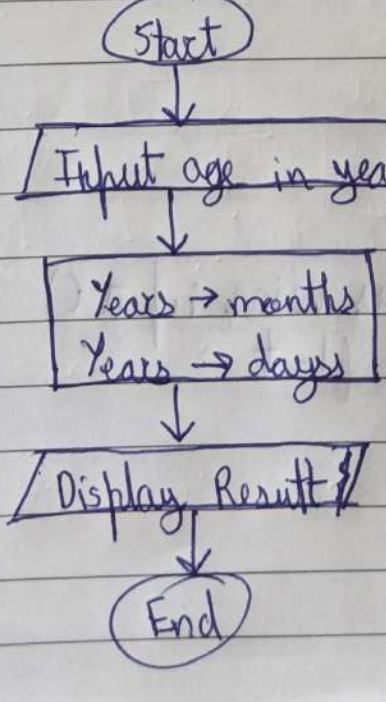


9. Write a program that prompts the user to enter their age and point out their age in year, months and days.

→
```
age = int (input ("Enter your age in years : "))
months = age * 12
days = age * 365
print ("Age in years:", age)
print ("Age in months :", months)
print ("Age in days : ", days)
```

* Algorithm :-
1. Start
2. Read age in years
3. Convert years into months and days
4. Display years, months, days
5. Stop.

* Flowchart :-



10. Write a program to demonstrate the concept of data type & conversion.

→ Code :

```
a=10
b = float (a)
c= str (a)
print (a , type (a))
print (b, type (b))
print (c , type (c))
```

* Algorithm :-
1. Start
2. Assign integer value.
3. Convert integer to float and string.
4. Display value and their data type.
5. End

* Flowchart :-